

Document reference	NPS / 002 / 025						
Version: -	1.0	Date of Issue: -	Oct-09	Page	1	of	8

NPS/002/025 – Technical Specification for Replacement Covers and Frames for Underground Link Box Chambers.

1. Purpose:

This document details the requirements for a range of replacement metal frames and covers for historic design of 2 and 4-way underground link box chambers currently in service on the distribution network of CE Electric UK.

This document supersedes the following documents, all copies of which should be destroyed.

Ref:	Version:	Title:
Nil	Nil	Nil

2. Scope:

Replacement metal frames and concrete filled and unfilled covers are required for situations where the cover or frame of an existing historic design brick built 2 and 4-way link box chamber requires changing due to mechanical damage and or general wear and tear. The covers and frames shall be suitable for use in footpaths and areas where slow moving vehicles including HGV vehicles could operate.

3. Technical Requirements:

Frames:

Metal frames manufactured from 6mm (Min) hot dipped galvanised steel to BS EN1461 are required to sit on top of the existing brick work. The frame shall be provided with a flange with slots holes to facilitate the bolting down of the frame to the existing brickwork.

All butt joint welds shall be flush and free from burrs.

The sides of the frame and the supplied concrete filled/unfilled covers shall be 'flush' with each other as to not cause a tripping hazard.

For larger opening chambers (4-way link boxes) where two x single covers are supplied, a removable cross member shall be incorporated into the frame to aid the removal of the covers. A minimum of 30mm overlap flange is required beneath each unsupported cover end.

Concrete in filled Cover:

Concrete filled covers shall meet the requirements of Group 2 class B125 as detailed in BS EN 124-1994 and be manufactured from 6mm (Min) hot dipped galvanised steel to BS EN1461 and supplied with reinforced high strength concrete infill.

The cover shall be supplied with centralised 'keyhole' lifting holes and have tapered sides to aid in the removal of the cover from the frame.

All covers shall be interchangeable with any other within a frame.

A centralised badge which states 'Danger Electricity' shall be embedded into the concrete and so recessed as to not cause a tripping hazard.

Document reference		NPS / 002 / 025					
Version: -	1.0	Date of Issue: -	Oct-09	Page	2	of	8

Unfilled Covers:

Unfilled covers are required for situations where the paving patterns are to be continued over the access point.

Covers shall meet the requirements of group 2 class B125 as detailed in BS EN 124-1994 and be manufactured from 6mm (Min) hot dipped galvanised steel to BS EN1461 and shall be capable of accommodating a paving thickness up to 80mm.

The cover shall be supplied with centralised 'keyhole' lifting holes and incorporate a slide out type feature to aid its removal from the frame.

Due to the possibility that rain water may become 'trapped' within the frame of the cover; a 20mm diameter (approx) hole shall be positioned within the base of the cover.

3.1 General Requirements:

Appendix 2 show the opening dimensions of two typical brick built chambers.

3.2 Type Test Requirements:

Type tests shall be in accordance with the appropriate British Standards; a self certification conformance declaration, appendix 1, shall be completed by the supplier.

Document reference		NPS / 002 / 025					
Version: -	1.0	Date of Issue: -	Oct-09	Page	3	of	8

4. References:

4.1 External Documentation

Reference	Title
BS EN 124 – 1994	Gully tops and manhole tops for vehicular and pedestrian areas. Design requirements, type testing, marking, quality control.
BS EN 1461 - 2009	Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods

4.2 Internal documentation

Reference	Title
Nil	

5. Definitions

Term	Definition
Nil	

Document reference	NPS / 002 / 025						
Version: -	1.0	Date of Issue: -	Oct-09	Page	4	of	8

6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
Richard McMahon	Standards Engineer		

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Mark Nicholson	Head of System Strategy		

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
Chris Dearn	Building & Civil Project Manager		

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator		

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessments Manager		

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement		

Document reference		NPS/ 002 / 000					
Version: -	1.0	Date of Issue: -	Sept-09	Page	5	of	8

Appendix 1 - SELF CERTIFICATION CONFORMANCE DECLARATION

Replacement metal frames and covers for historic design of 2 and 4-way underground link box chambers covered by BS EN1461 and BS EN 124 shall comply with the latest issues of the relevant international and international standards.

BS EN1461 and BS EN124 are intended to amplify and/or clarify the requirements of those Standards.

This check sheet identifies the clauses of the aforementioned standard relevant to replacement metal frames and covers for historic design of 2 and 4-way underground link box chambers The manufacturer shall declare conformance or otherwise, clause by clause, using the following levels of conformance declaration codes.

Conformance declaration codes

N/A = Clause is not applicable/ appropriate to the product.

Cs1 = the product conforms fully with the requirements of this clause.

Cs2 = the product conforms partially with the requirements of this clause.

Cs3 = the product does not conform to the requirements of this clause.

Cs4 = the product does not currently conform to the requirements of this clause, but the manufacturer proposes to modify and test the product in order to conform.

Instructions for completion

- When Cs1 code is entered no remark is necessary
- When any other code is entered the reason for non- conformance shall be entered
- Prefix each remark with the relevant 'BS EN' 'IEC' or 'ENATS' as appropriate.

Manufacturer:

Product Reference:

Name:

Signature:

Date:

NOTE: One sheet shall be completed for each item or variant submitted.

Document reference		NPS/ 002 / 000					
Version: -	1.0	Date of Issue: -	Sept-09	Page	6	of	8

Appendix 1 - Cont'

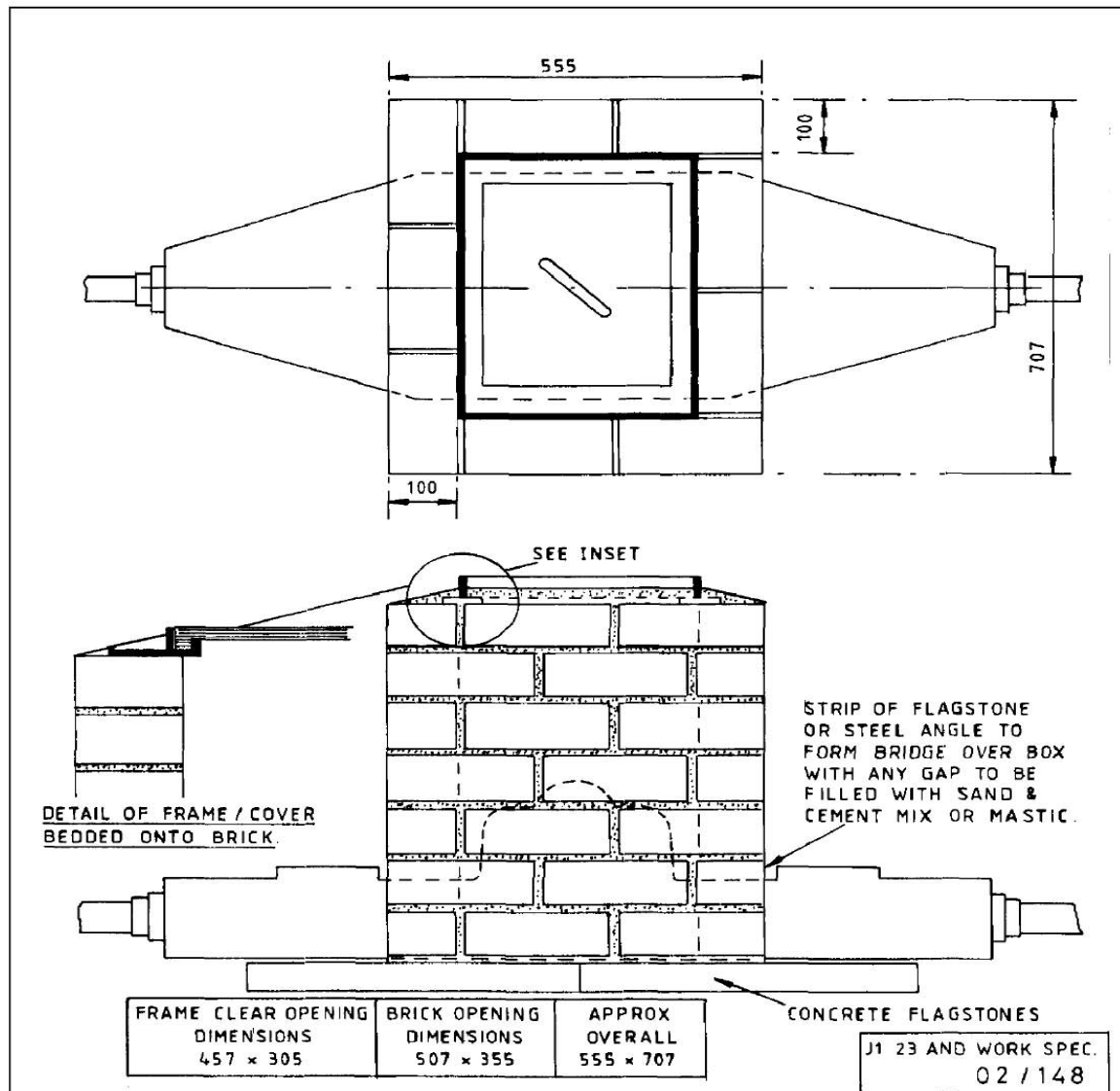
Gully tops and manhole tops for vehicular and pedestrian areas – Design requirements, type testing, marking and quality control.

BS EN 124:1994.			
Clause/Sub-clause	Requirement	Conformance Code	Remarks
5	Place of Installation: Group 2 (Min. class B 125)		
6. 6.1.1 - Table 1	Galvanizing coating of rolled steel		
6.2	Minimum compressive strength of concrete		
9	Markings		

CE ELECTRIC UK

Document reference	NPS/ 002 / 000						
Version: -	1.0	Date of Issue: -	Sept-09	Page	7	of	8

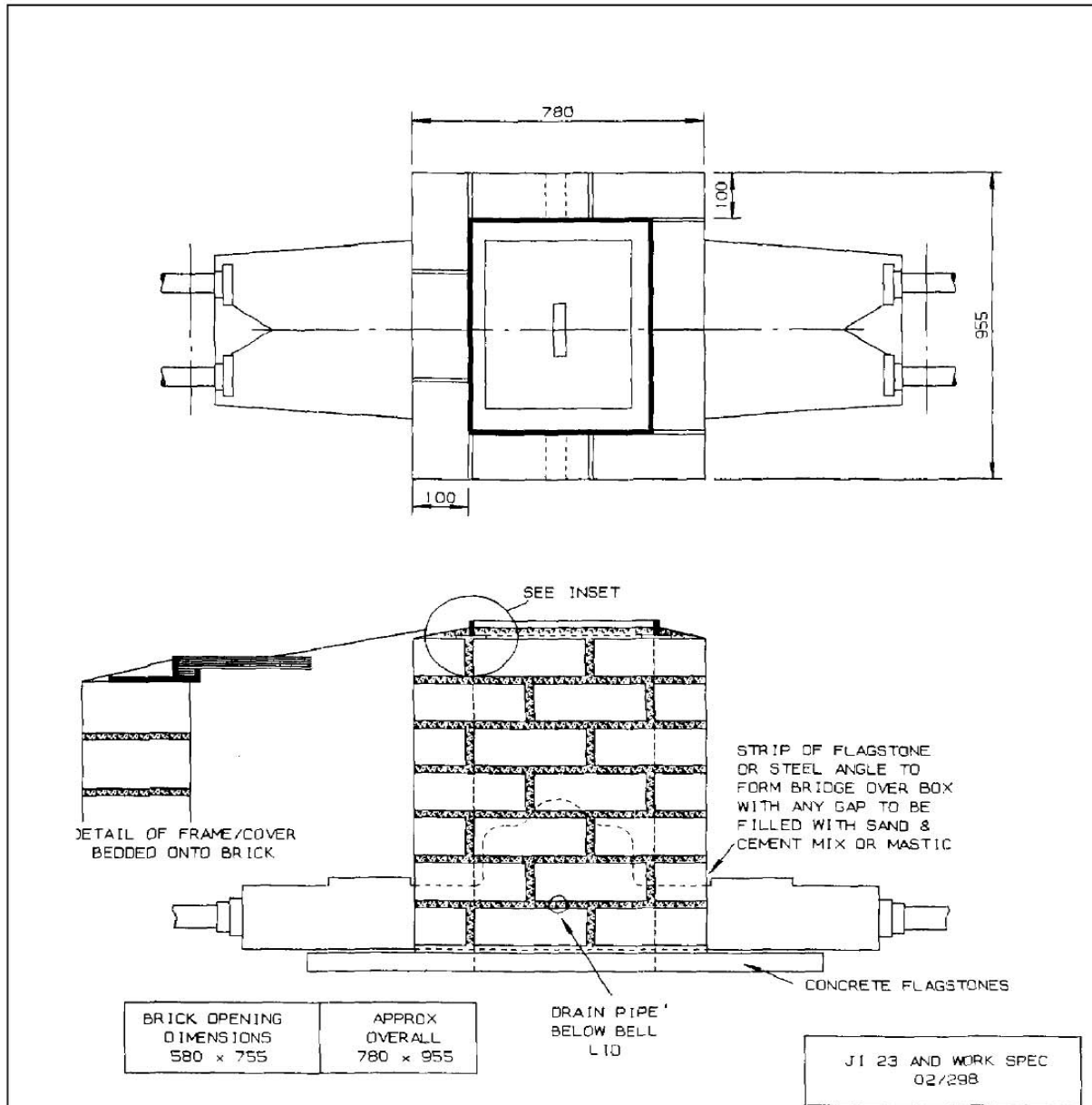
Appendix 2



CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF	
Manufacturer Details		PAVEMENT PIT (BRICK BUILT) FOR LV UNDERGROUND 2 WAY LINK DISCONNECTING BOX (RESIN FILLED)	
Sheet No. 2	Scale N.T.S.	Type LEGACY	OTHER
Prepared By Barbara Gordon	Grid Reference	Ref No. C980144	Historic Drg.No. 1181040060
Revised 13/10/09	Checked By R MCMAHON	Revision A	Notes
Date Issued 13/10/09			

CAUTION! - This document may be out of date if printed

Document reference	NPS/ 002 / 000						
Version: -	1.0	Date of Issue: -	Sept-09	Page	8	of	8



 CE Electric UK		Lloyds Court, 78 Grey Street, Newcastle Upon Tyne, NE1 6AF			
Manufacturer Details		PAVEMENT PIT(BRICK BUILT) FOR LV UNDERGROUND 4 WAY LINK DISCONNECTING BOX (RESIN FILLED)			
Sheet No. 3	Scale N.T.S.				
Prepared By Barbara.Gordon		Type LEGACYOTHER			
Revised 13/10/09	Grid Reference	Ref No.	C980145	Historic Drg.No.	1181040060
Date Issued 13/10/09	Checked By R MCMAHON	Revision	A	Notes	